

8.1 Stationary Waves

Question Paper

Course	CIEA Level Physics
Section	8. Superposition
Topic	8.1 Stationary Waves
Difficulty	Easy

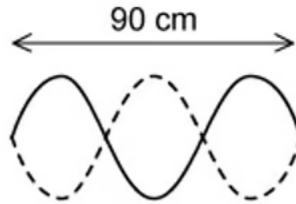
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A stationary wave shown in the diagram was created on a stretched string, it had two instants of maximum vertical displacement.



The frequency of the wave is 13 Hz.

What is the speed of the wave?

- A. 3.9 m s^{-1}
- B. 7.8 m s^{-1}
- C. 390 m s^{-1}
- D. 780 m s^{-1}

[1 mark]

Question 2

The table shows some statements about stationary and progressive waves.

Which row is correct?

	Progressive wave	Stationary wave
A	particles one wavelength apart vibrate in phase	particles one wavelength apart vibrate in phase
B	particles vibrate in phase with their immediate neighbours	particles in adjacent loops vibrate in antiphase
C	energy is transferred along the wave	energy is transferred along the wave
D	all particles vibrate with the same amplitude	all particles vibrate with the same amplitude

[1 mark]

Question 3

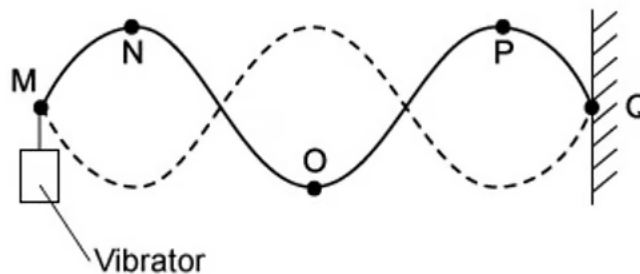
When a stationary wave is produced in a tube, which of the following statements describes the wave speed?

- A. it is the speed of a particle at a node
- B. it is the speed of one of the progressive waves that are producing the stationary wave
- C. it is the speed at which energy is transferred from one antinode to an adjacent antinode
- D. it is the distance between two adjacent nodes divided by the period of the wave

[1 mark]

Question 4

A stretched string is set up to demonstrate a stationary wave between two points M and Q.



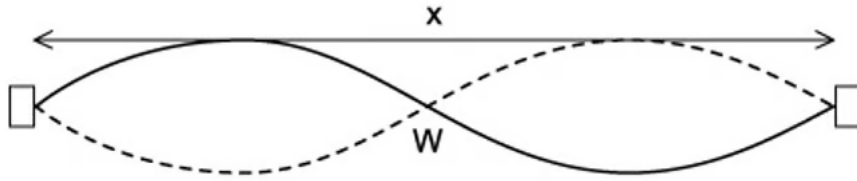
Which of the following statements are correct?

- A. the distance between M and Q is three wavelengths
- B. the wave shown has the lowest possible frequency
- C. points N and P vibrate in phase
- D. point O is at a node

[1 mark]

Question 5

A stretched string is used to demonstrate a stationary wave, as shown in the diagram.



Which row in the table correctly describes the length of x and the name of W ?

	Length x	Point W
A	two wavelengths	node
B	one wavelength	node
C	two wavelengths	antinode
D	one wavelength	antinode

[1 mark]

Question 6

A stationary wave on a string is shown in the diagram below. The standing wave has three nodes Q_1 , Q_2 and Q_3



Which of the following statements are correct ?

- A. all points on the string vibrate in phase
- B. all points on the string vibrate with the same amplitude
- C. points equidistant from Q_2 vibrate with the same frequency and the same amplitude
- D. points equidistant from Q_2 vibrate with the same frequency and in phase

[1 mark]

Question 7

A student formed a stationary wave in a tube by blowing over the top as shown in the diagram



Which of the following statements are correct ?

- A. the stationary wave in the cylinder is polarised
- B. the stationary wave will have an antinode at the bottom of the cylinder
- C. the fundamental frequency of the stationary wave decreases when some water is added to the cylinder
- D. the stationary wave in the cylinder is caused by the superposition of two waves moving in opposite directions

[1 mark]

Question 8

A stationary wave is formed when two waves travel in opposite directions through the same space.

Which of the following statements must also be correct?

- A. the waves must have equal speed, frequency and wavelength
- B. the waves must have equal frequency, but a different speed and wavelength
- C. the waves must have equal wavelength, but a different speed and frequency
- D. the waves must have equal speed, but a different wavelength and frequency

[1 mark]

Question 9

A student sets up a sound wave in a long tube, and the tube is closed at one end. The tube length is changed until the sound from the tube is loudest.

Which of the following statements describe the nature of the wave in the tube?

- A. transverse and stationary
- B. transverse and progressive
- C. longitudinal and stationary
- D. longitudinal and progressive

[1 mark]

Question 10

A student wants to demonstrate a stationary wave.

What could they use to produce this?

- A. passing water waves through a narrow slit
- B. passing monochromatic light through a double slit
- C. making a loud sound near a mountain
- D. blowing air over the top of an empty bottle

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Wave speed = frequency $\times$ wavelength
 - $v = f \lambda$
- Frequency = 13 Hz
- Wavelength: in the diagram there are 1.5 waves in 90 cm
 - $\frac{90}{1.5} = 60 \text{ cm} = 0.6 \text{ m}$
- Therefore, $v = 13 \times 0.6 = 7.8 \text{ m s}^{-1}$

2

The correct answer is **A** because:

- In a progressive wave, the wavefronts move through the medium transferring energy from one point to the next
- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
 - The waves in a standing wave are one wavelength apart and vibrate in phase
 - Particles in a progressive wave can also be one wavelength apart and vibrate in phase

B is incorrect because in a standing wave, adjacent particles are vibrating in phase with each other

C is incorrect because only a progressive wave transfers energy, in a standing wave there is no net transfer of energy

D is incorrect because in a standing wave adjacent points each oscillate with a different amplitude

3

The correct answer is **B** because:

- In a progressive wave, the wavefronts move through the medium transferring energy from one point to the next
- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The wave speed of a standing wave is the speed that one of the progressive wave travels that is producing the standing wave

A is incorrect because the wave speed is, $v = f \lambda$, at a node the wave speed = 0

C is incorrect because a standing wave does not transfer energy

D is incorrect because the distance between two antinodes would be the wavelength of the wave. The period of the wave is the time it takes for one wave to travel

4

The correct answer is **C** because:

- N and P are vibrating in phase
 - This means they are vibrating up and down at the same time

A is incorrect because there are 1.5 wavelengths between M and Q

B is incorrect because a wave with the lowest frequency will be half a wave

D is incorrect because point O is at an antinode

5

The correct answer is **B** because:

- A wavelength is the length of one complete wave
 - Each loop is the equivalent of one-half of a wavelength
- A node is the point on a standing wave of minimum amplitude

A & C are incorrect because there are not two wavelengths in the diagram

D is incorrect because the antinode is the point of maximum amplitude of the wave

and found at the crest of the loop

6

The correct answer is **C** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The points that are equidistant from Q_2 will vibrate with the same frequency and amplitude

A is incorrect because not all the points on the wave vibrate in phase. Points within the loops will vibrate in phase but vibrate out of phase with the adjacent loop

B is incorrect because not all the points on the string vibrate with the same amplitude. Points closer to the nodes have a smaller amplitude than those as the crest of the loop

D is incorrect because points that are equidistant from Q_2 are not in phase as points in adjacent loops are out of phase

7

The correct answer is **D** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- A standing wave is one in which two waves combine constructively giving an amplitude of twice the original wave
 - Therefore, the stationary wave in the cylinder is caused by the superposition of two waves moving in opposite directions

A is incorrect because sound waves can't be polarised

B is incorrect because the waves will meet at a node at the fixed end of the cylinder

C is incorrect because decreasing the length of the column (done by adding water to the cylinder) will increase the fundamental frequency of the system

8

The correct answer is **A** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- A standing wave is one in which two waves combine constructively giving an amplitude of twice the original wave
 - Therefore, the waves must have equal speed, frequency and wavelength

B & D are incorrect because if the waves had a different wavelengths, then they would not be in phase

C is incorrect because the waves must have the same frequency to create a standing wave

9

The correct answer is **C** because:

- The wave is a sound wave, which is a longitudinal wave
 - Longitudinal waves travel parallel to the direction of travel
- The wave created by constructive combining of the sound waves created the largest amplitude, so the loudest sound
 - Hence the wave created is a stationary wave

A & B are incorrect because sound waves are not transverse waves; transverse waves travel perpendicular to the direction of travel

D is incorrect because superposition of the wave has occurred creating a standing wave, so the wave is not progressive

10

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10

The correct answer is **D** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- When air is blown over the top of an empty bottle, the bottom of the bottle has a fixed end, creating a wave travelling in the opposite direction with the same frequency and wavelength

A is incorrect because a single slit would diffract the wave into semicircular waves

B is incorrect because this would create the characteristic fringes caused by the superposition of the overlapping light from the two slits

C is incorrect because this would create an echo, which is a reflection of a sound wave

